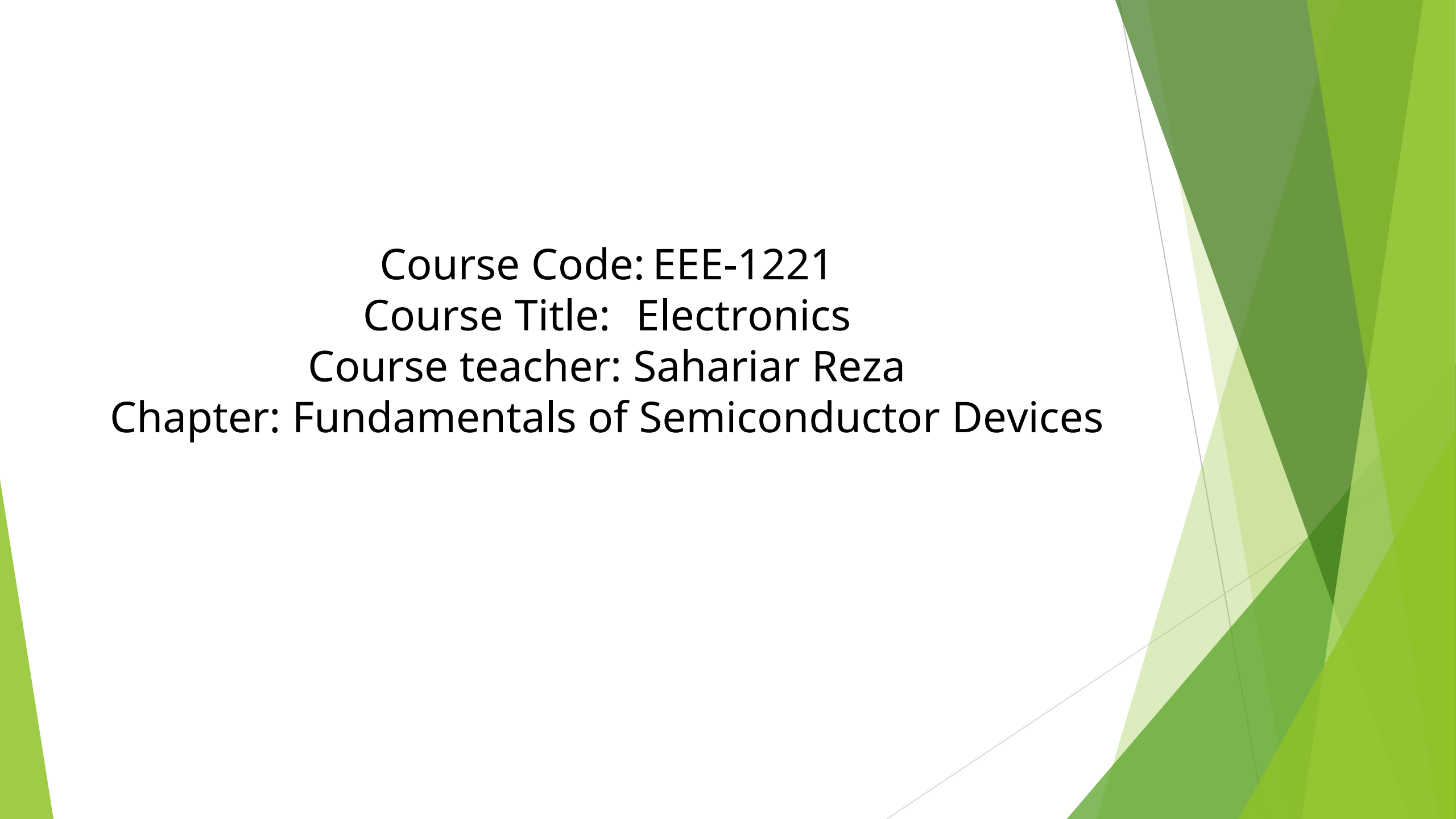
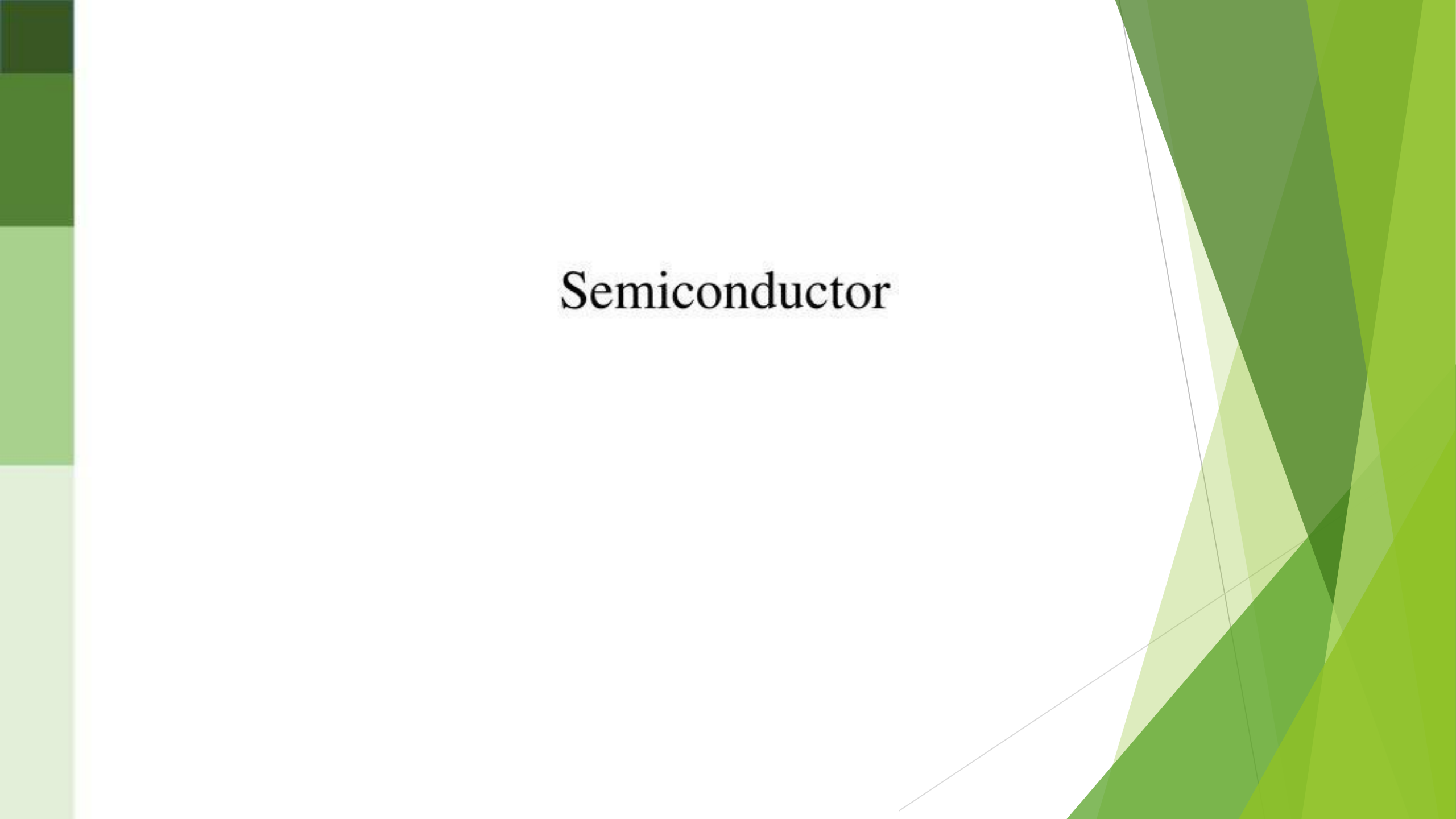


The background features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern, layered effect on the right side of the slide.

Course Code: EEE-1221
Course Title: Electronics
Course teacher: Sahariar Reza
Chapter: Fundamentals of Semiconductor Devices

OUTLINES

- ❖ Semiconductor & its Properties
- ❖ Classification of semiconductor (**Intrinsic & Extrinsic semiconductors**)
- ❖ N-type & P-type semiconductor
- ❖ PN junction (formation or biasing).
- ❖ Forward bias & Reverse bias...
- ❖ J_V-I Characteristics
- ❖ Definition of diode



Semiconductor

Classification of materials

On the basis of electrical conductivity, materials can be classified into three types.

- ❖ Conductor
- ❖ Semiconductor (not good conductor but in between conductor and insulator)
- ❖ Insulator

“Semiconductor can carry current under some condition”

Semiconductor

Substance	Nature	Resistivity
Copper	Good conductor	$1.78 \times 10^{-8} \Omega m$
Germanium	Semiconductor	$0.6 \Omega m$
Glass	Insulator	$9 \times 10^{11} \Omega m$

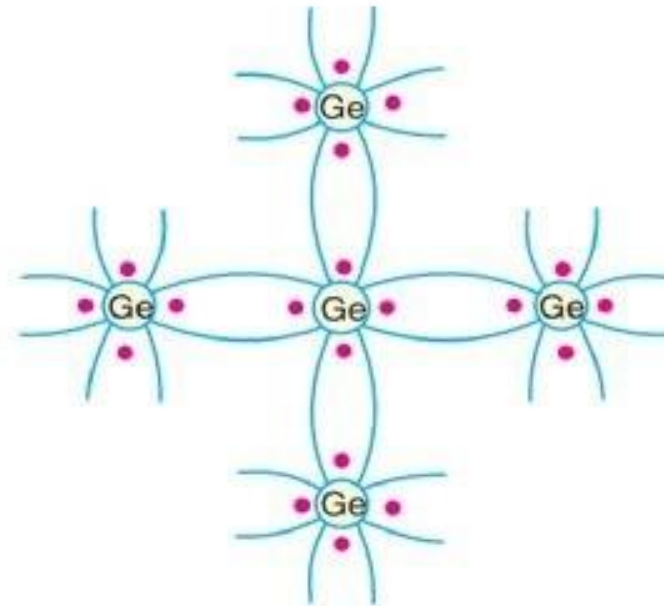
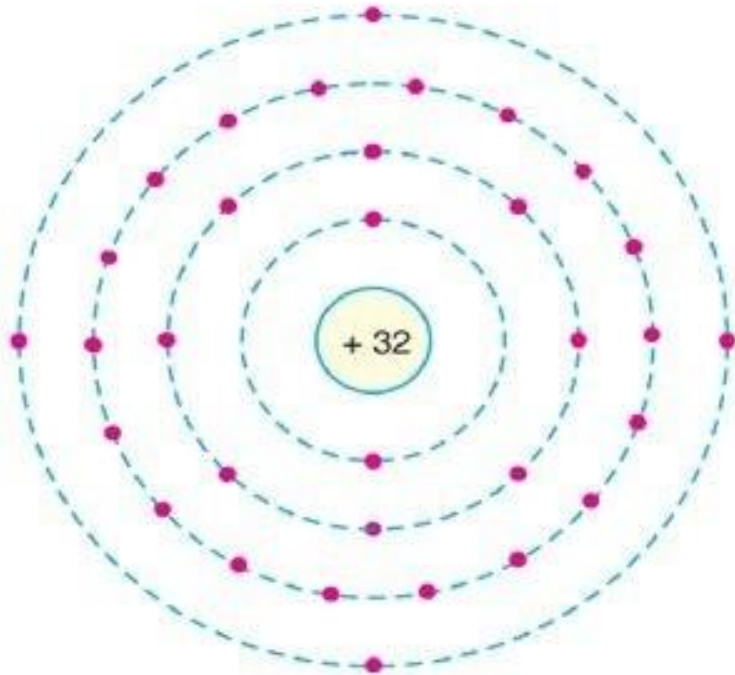
“A semiconductor is a substance whose resistivity lies in between conductor and insulator”(according to electrical conductivity)

Properties of semiconductor

- The resistivity of a semiconductor is less than an insulator but more than a conductor.
- Semiconductors have negative temperature co-efficient of resistance i.e. the resistance of a semiconductor decreases with the increase in temperature and vice-versa. For example, germanium is actually an insulator at low temperatures but it becomes a good conductor at high temperatures.
- When a suitable metallic impurity (e.g. arsenic, gallium etc.) is added to a semiconductor, its current conducting properties change appreciably.

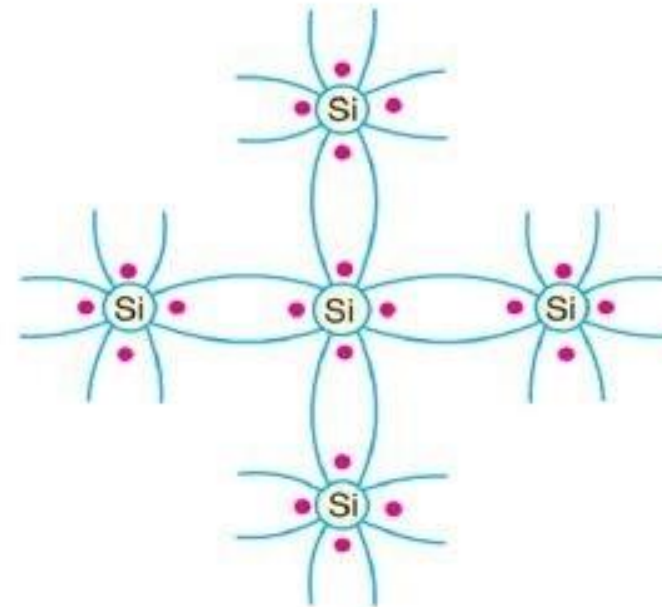
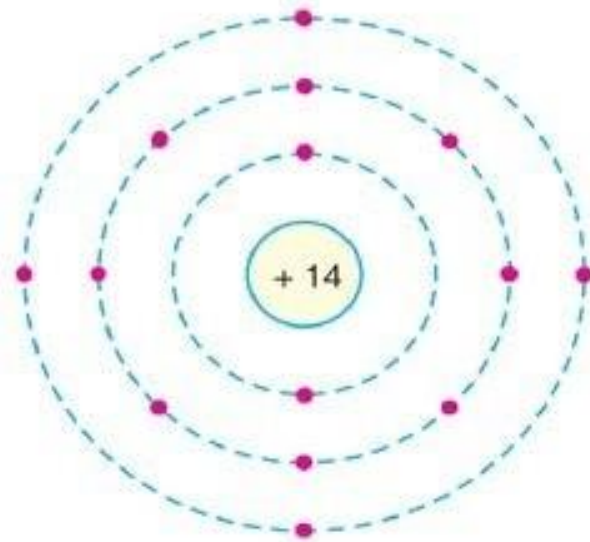
Commonly used semiconductors

Germanium (32)

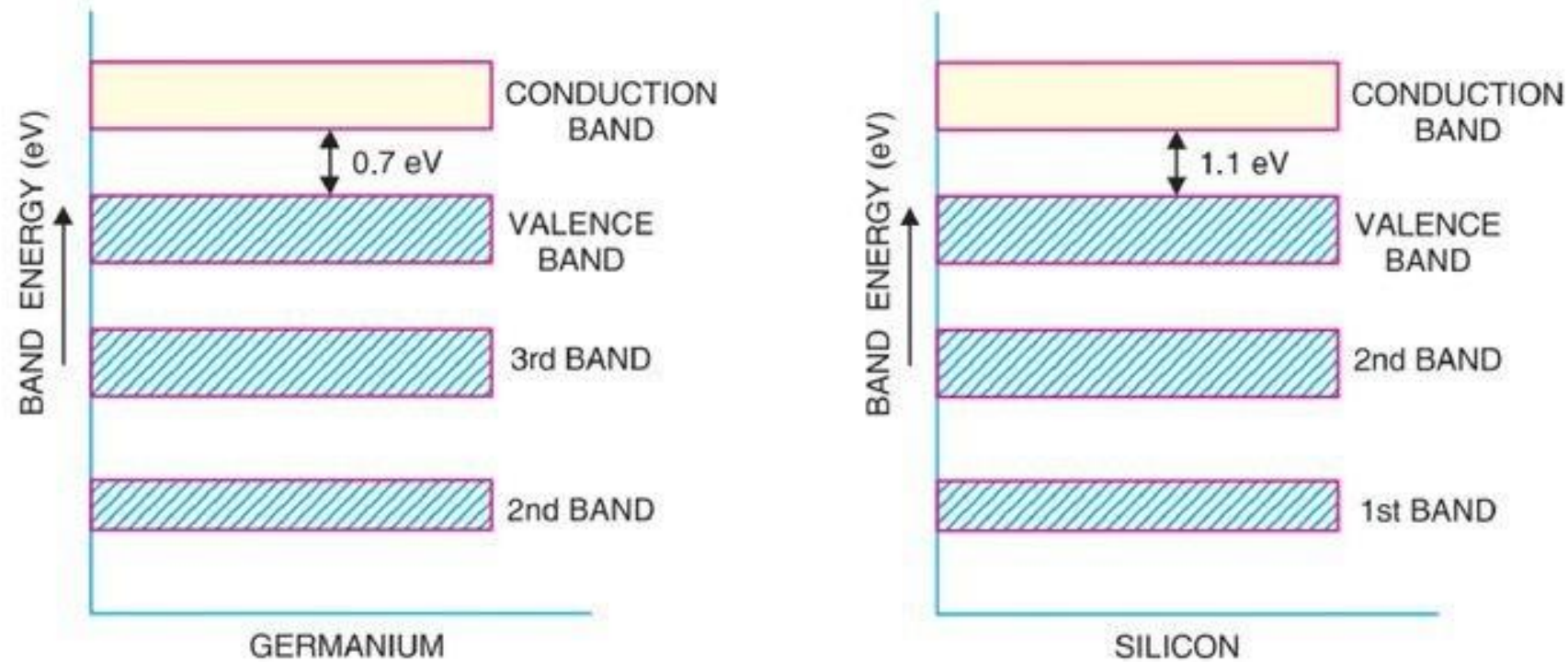


Commonly used semiconductors

Silicon (14)

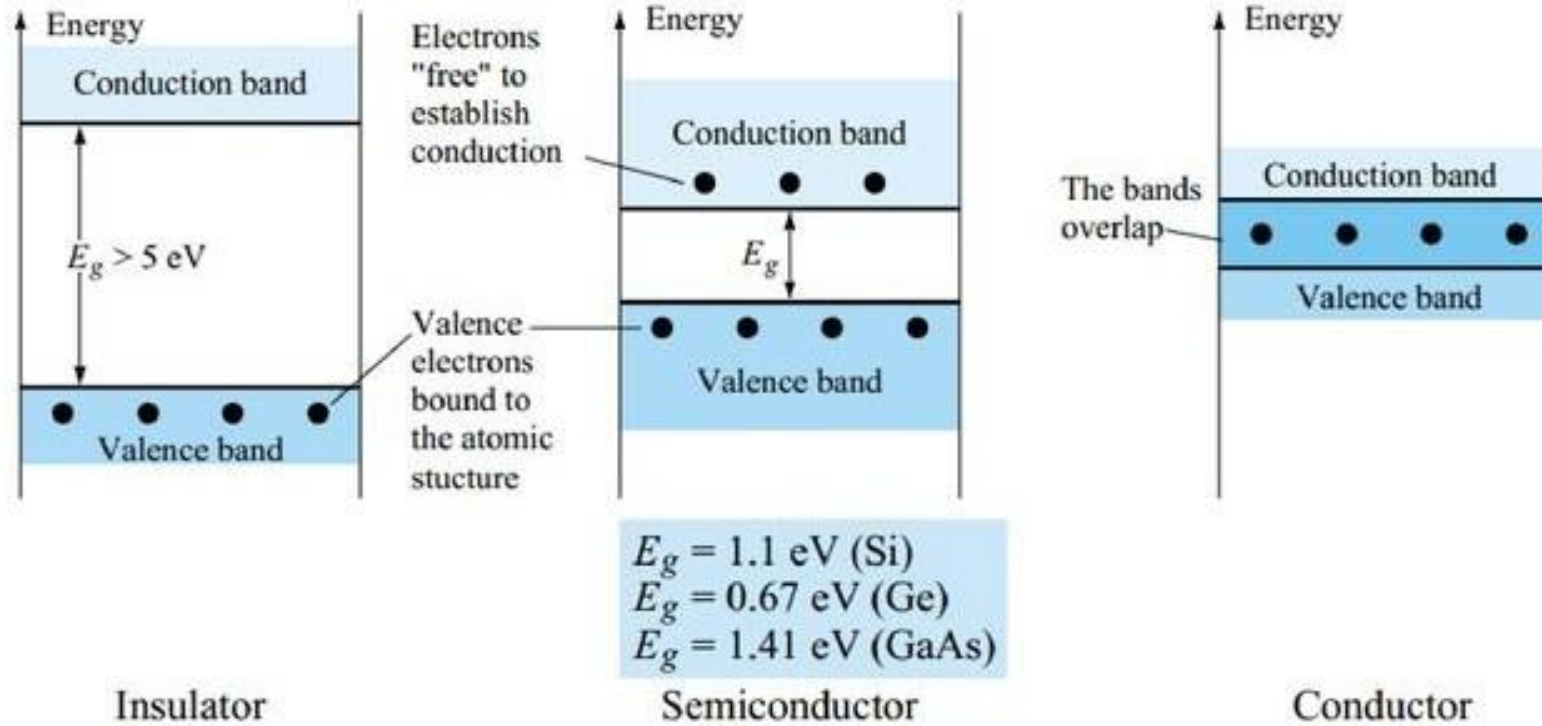


Properties of semiconductor



A semiconductor is a substance which has almost filled valence band and nearly empty conduction band with a very small energy gap (1.1 eV) separating the two.

Energy band diagram



Classification of semiconductor

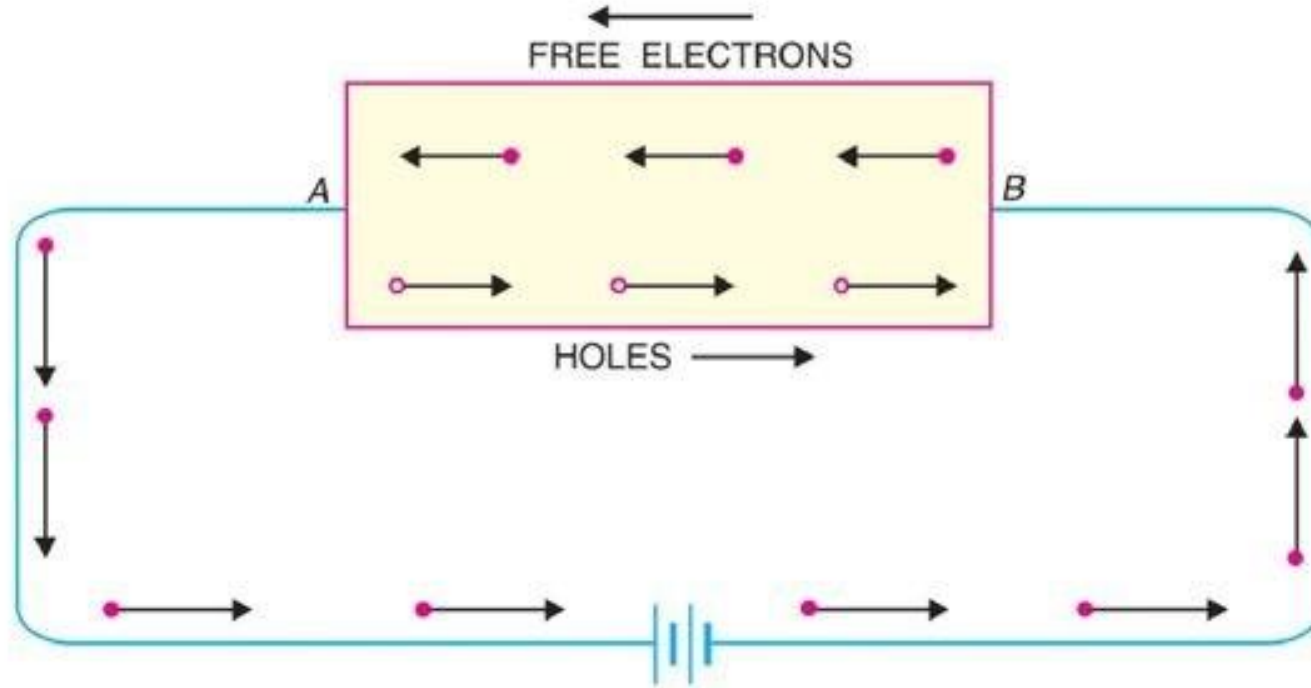
- ❖ Intrinsic semiconductor
- ❖ Extrinsic semiconductor (this can be classified into two types)
 - ✓ n-type semiconductor
 - ✓ p-type semiconductor

Intrinsic semiconductor: An extremely pure form of semiconductor is called intrinsic semiconductor.

Intrinsic semiconductor

❖ Intrinsic semiconductor

An extremely pure form of semiconductor is called intrinsic semiconductor.



Why we need extrinsic semiconductor?

Because the conduction capability of intrinsic semiconductor is very small.

- ❖ But for useful purposes in semiconductor device we need to increase the amount of current.
- ❖ That's means we need to increase the conductivity of the material
- ❖ Increase in conductivity means increase in carrier concentration.

Carrier concentration can be altered in a semiconductor by adding impurity

“The process of impurity addition to semiconductor is called doping”

Types of doping

Two types

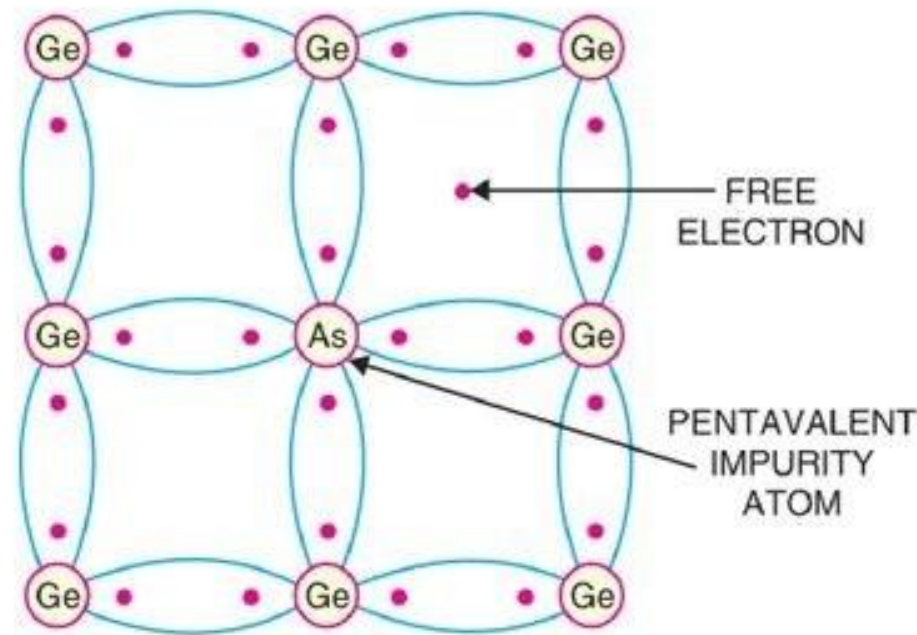
- I. n- type doping (n- negative type)
- II. p- type doping (p- positive type)

“Extrinsic semiconductor is obtained from the process of impurity adding to the intrinsic semiconductor to change the carrier concentration to change the electronic property of the semiconductor material, this process is named as doping”

n-type doping

From n-type doping we obtain n-type semiconductor

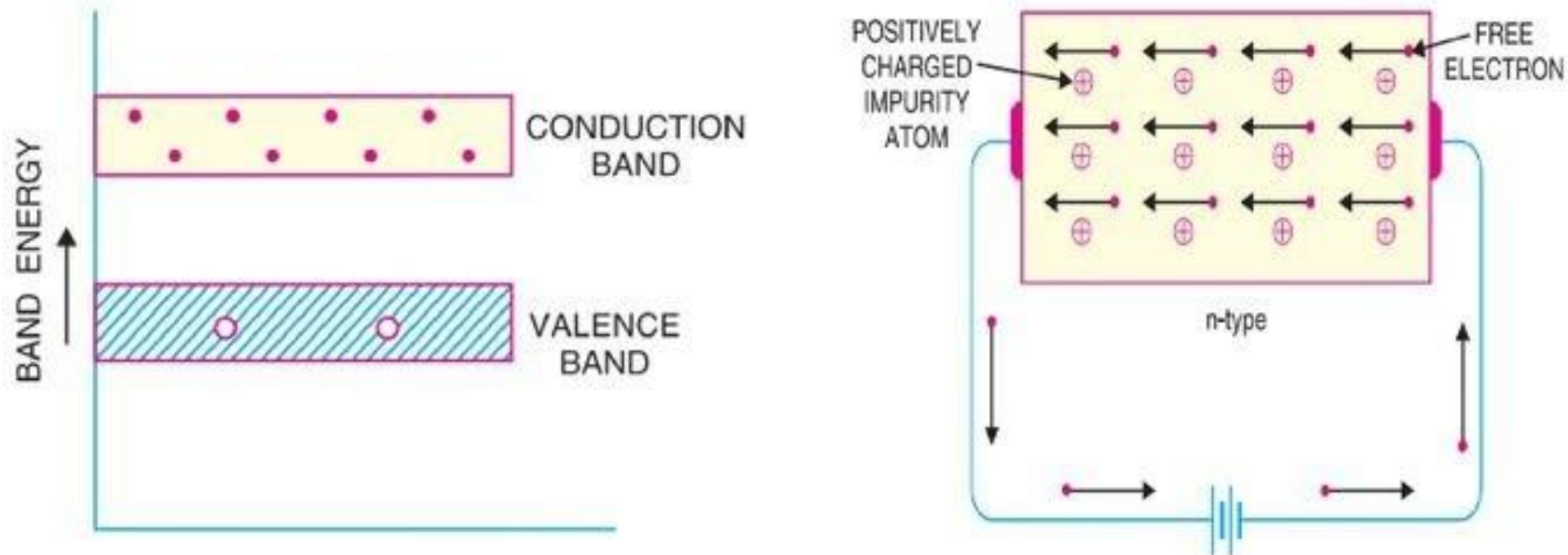
“When a small amount of pentavalent impurity is added to a pure semiconductor, it is known as n-type semiconductor”



Impurities which produce n-type semiconductor are known as donor impurities because they donate or provide free electrons to the semiconductor crystal.

n-type doping

The current conduction in an n-type semiconductor is predominantly by free electrons i.e. negative charges and is called n-type or electron type conductivity

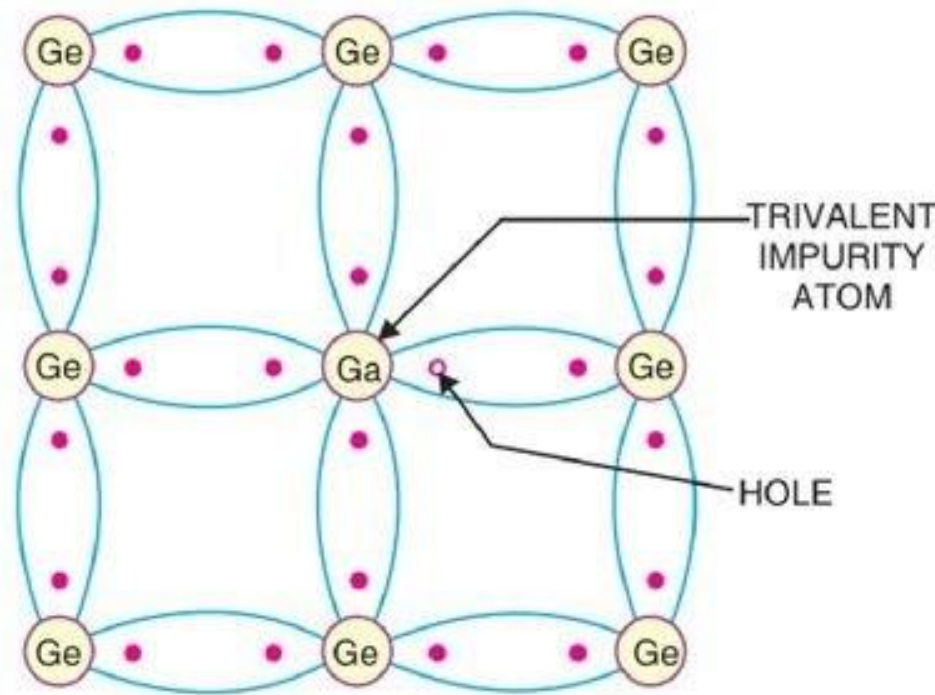


“So, majority carrier in n- type semiconductor is electron and minority carrier is hole”

p-type doping

From p-type doping we obtain p-type semiconductor

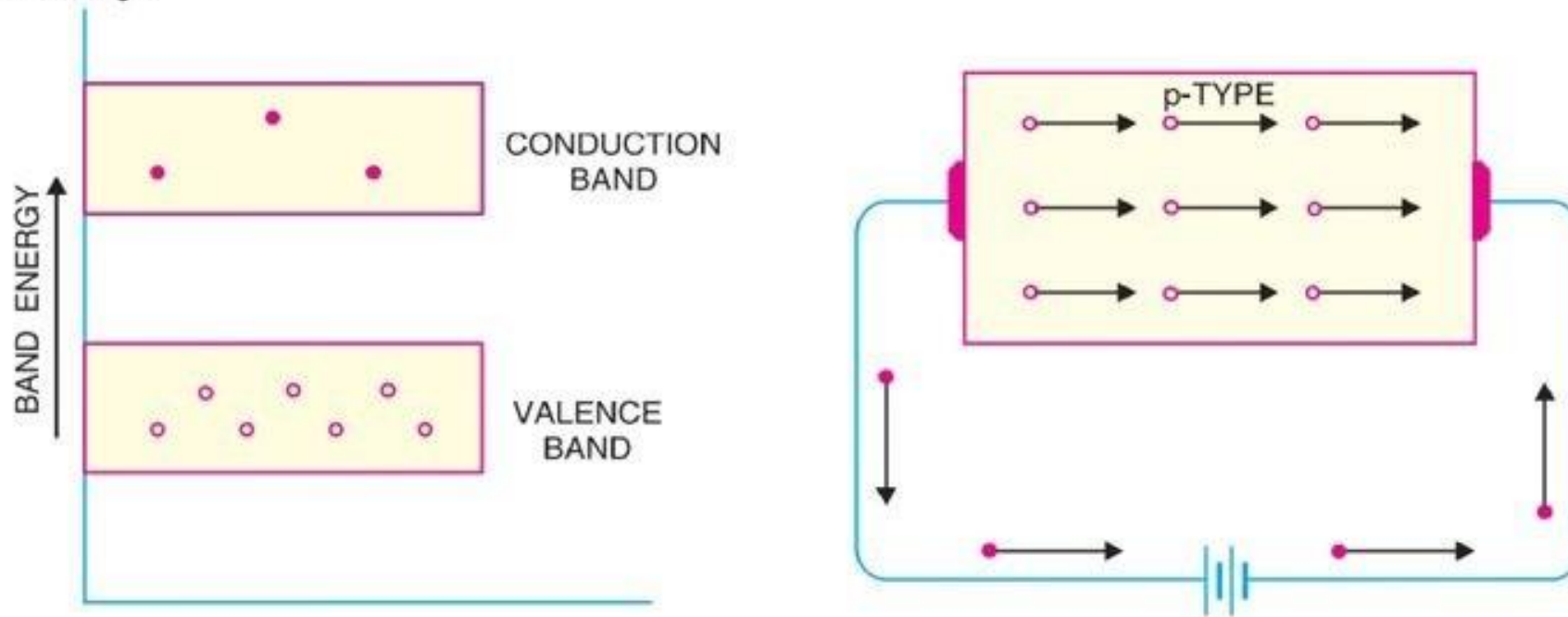
“When a small amount of trivalent impurity (In, Ga) is added to a pure semiconductor, it is called p-type semiconductor.”



Such impurities which produce p-type semiconductor are known as acceptor impurities because the holes created can accept the electrons.

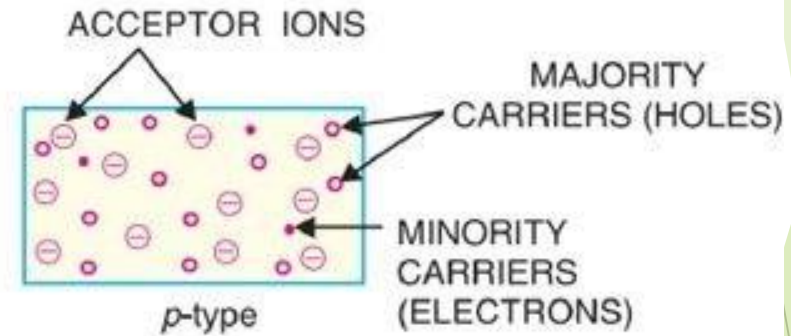
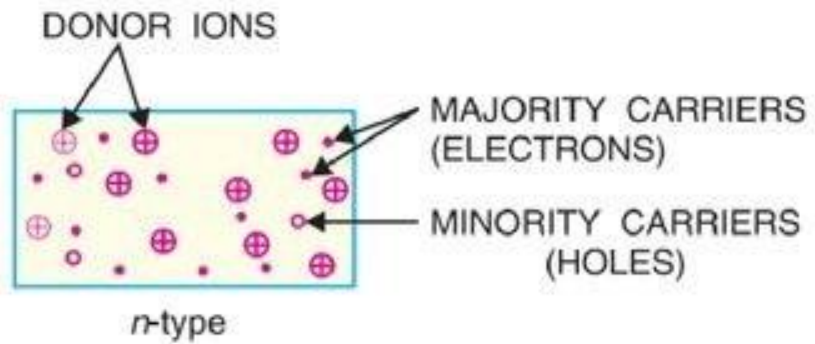
p- type doping

The current conduction in an p-type semiconductor is predominantly by free electrons i.e. negative charges and is called n-type or electron type conductivity.



“Majority carrier in p- type semiconductor is hole and minority carrier is electron”

Majority and minority carrier



Semiconductor devices
(crystal diode, transistors)

**semiconductor device is nothing but a simple p-n
junction or many p-n junctions**

p-n junction

When a pure semiconductor's one side is doped with n-type doping and another side is doped with p-type doping a junction is formed between the n-type and p-type semiconductor is called p-n junction

“p-n junction is important since the control element for semiconductor devices.”

n-type semiconductor – majority carrier(electron), minority carrier (Hole)

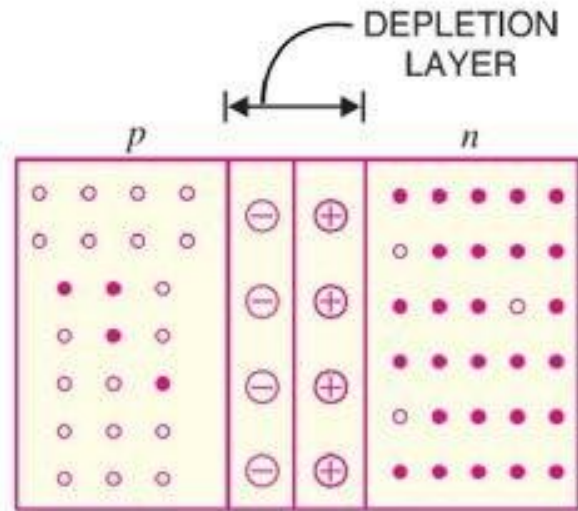
p- type semiconductor- majority carrier(hole), minority carrier (electron)

The density of electrons and holes are high in n side and p side respectively

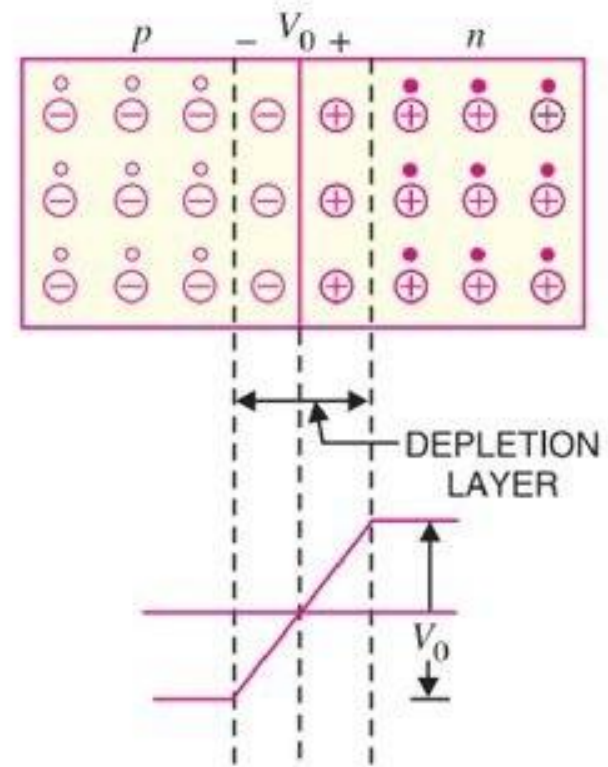
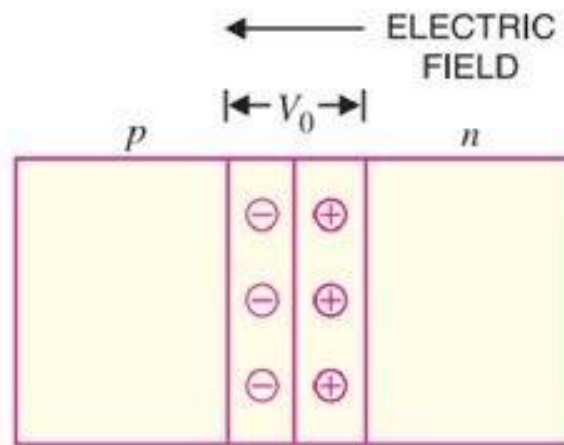
p-n junction

- The density of electrons and holes are high in n side and p side respectively.
- So electrons are diffused from n side and move towards p side and holes are diffused from p side and move towards n side.
- These electrons and holes recombine near the junction
- Since the density of electrons is decreased at n side, so positive charges appears near n side at the junction.
- And similarly density of holes is decreased at p side, so negative charges appears near the p side at the junction.
- This region of positive and negative charges at the junction is called depletion region. (**depleted– emptied, no free charge carrier in this region**)

p-n junction



(i)



p-n junction

- ❖ At equilibrium the depletion region acts as a barrier for further diffusion of electron and holes, stops the movement of electrons and holes.
- ❖ The positive and negative charges at this region set up electric field.
- ❖ Direction of electric field is from positive charge to negative charge
- ❖ The potential difference in the depletion region is called, Built in potential or barrier potential.

The value of built in potential, v_o in a Si p-n junction is 0.7 v

And the value of built in potential, v_o in a Ge p-n junction is 0.3 v

p-n junction

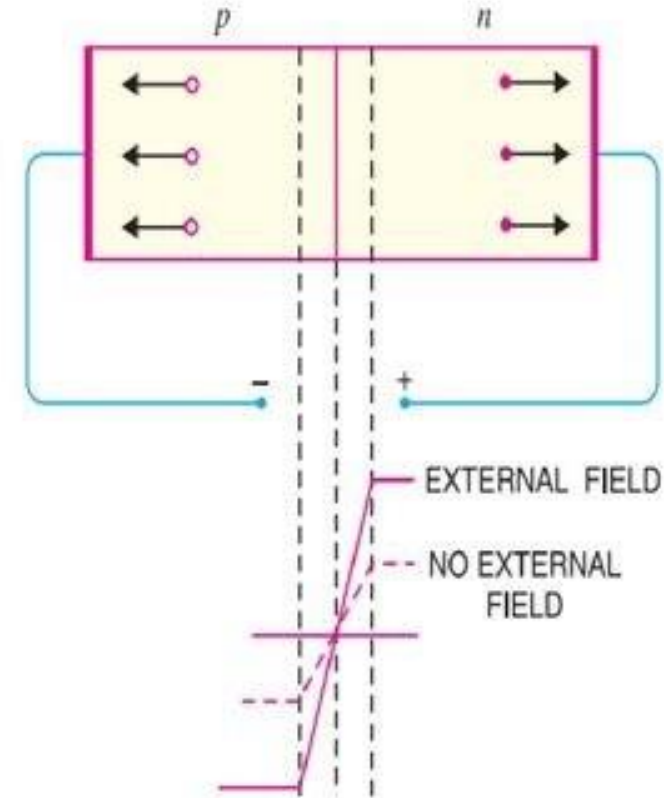
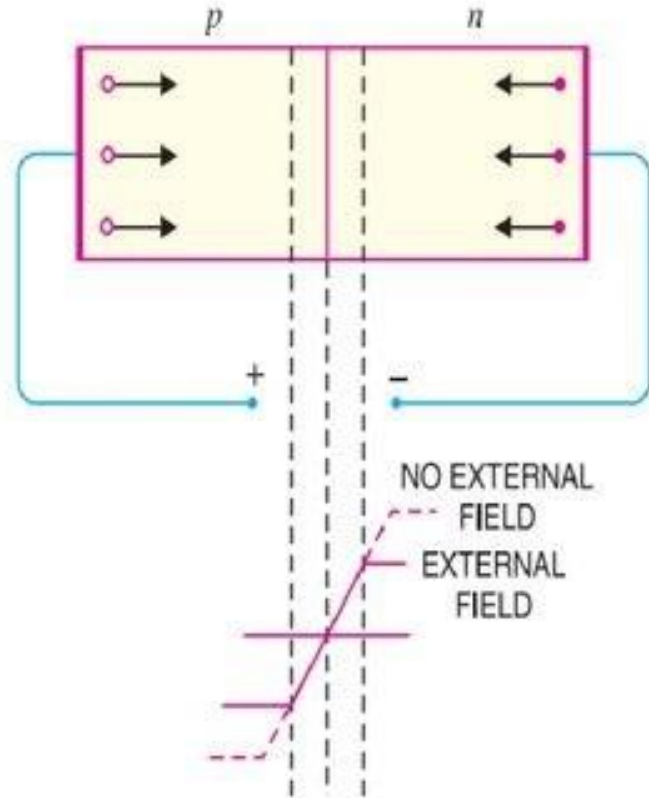
In electronics, the term bias refers to the use of d.c voltage to establish certain operating conditions.

Two types of biasing

- Forward bias
- Reverse bias

Forward Biasing

When external d.c. voltage applied to the junction is in such a direction that it cancels the potential barrier, thus permitting current flow, it is called forward biasing.



Reverse Biasing

When the external d.c. voltage applied to the junction is in such a direction that potential barrier is increased, it is called reverse biasing.

“In forward biasing the junction resistance decreases and in reverse biasing resistance increases”

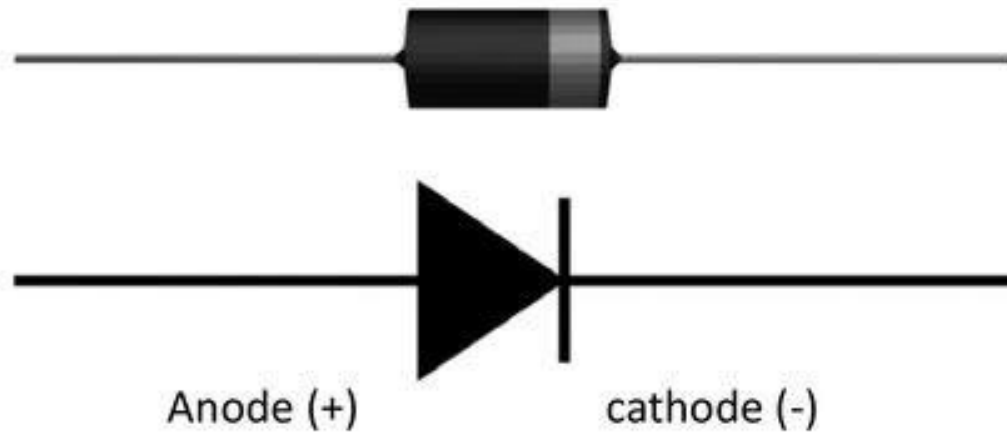
Breakdown voltage: It is the minimum reverse voltage at which pn junction breaks down with sudden rise in reverse current.

Peak inverse voltage: It is the maximum reverse voltage that can be applied to the p-n junction without damage to the junction.

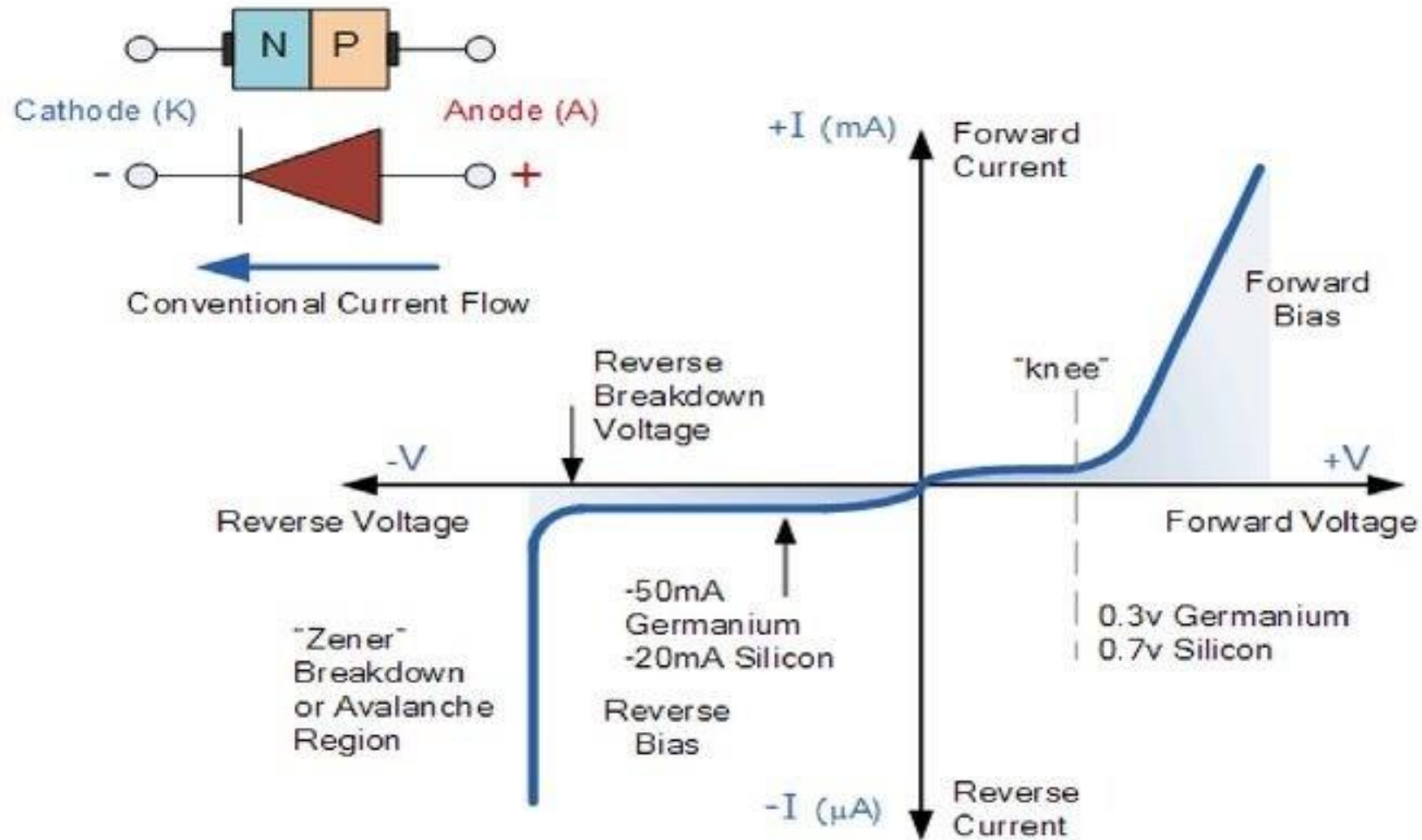
$PIV < \text{Breakdown voltage}$

Crystal diode

A pn junction is known as a semi-conductor or *crystal diode.



V-I characteristics of a p-n junction diode (crystal diode)



V-I characteristics of a p-n junction diode (crystal diode)

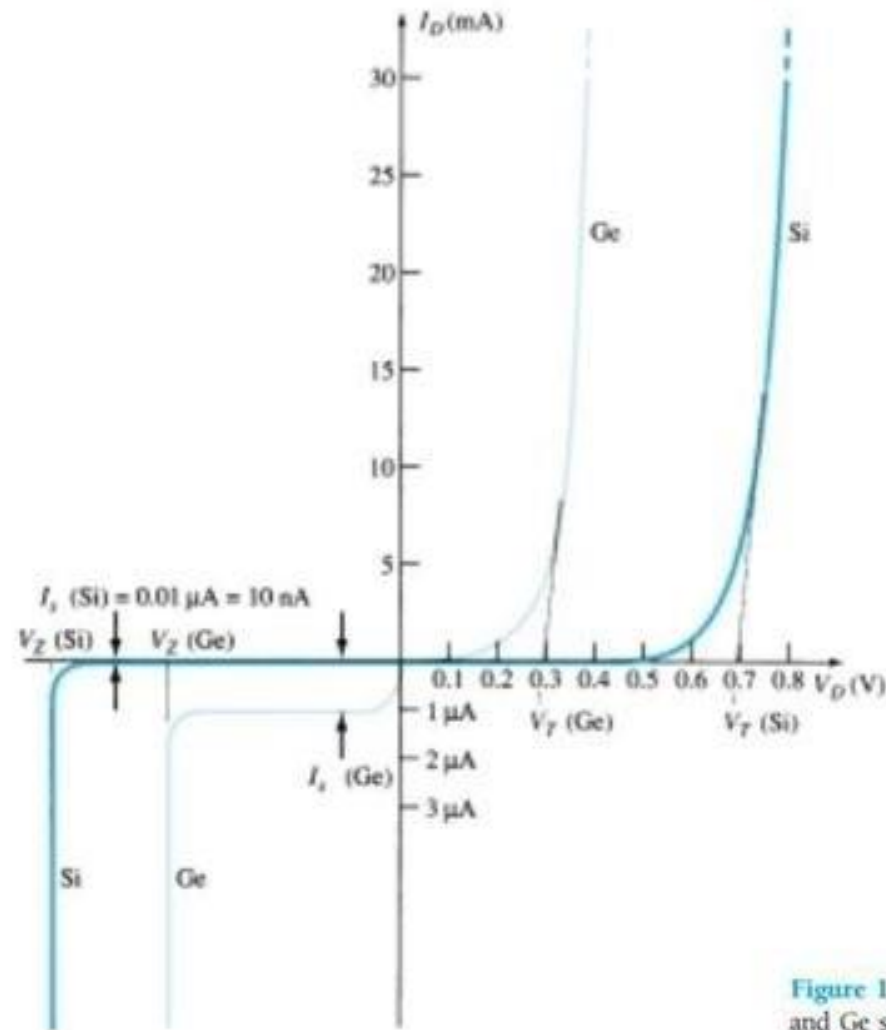


Figure 1.23 Comparison of Si and Ge semiconductor diodes.